Name: _____ Class: _____ Date: _____ ID: A

Grade 11 Pre-Calculus: Chapter 1

Multiple Choice
Identify the choice that best completes the statement or answers the question.

____ **1.** (1 point) Which of the following numbers occurs in the sequence –12, –8, –4, 0, 4, . . .?
A 6 **C** 24
B –3 **D** 15

____ **2.** (1 point) The common difference in the arithmetic sequence 2, –6, –14, –22, . . . is
A –8 **C** –16
B –3 **D** 8

____ **3.** (1 point) The common difference in the arithmetic sequence $\frac{1}{2}, \frac{5}{6}, \frac{7}{6}, \frac{3}{2}, \frac{11}{6}, \ldots$ is
A $\frac{5}{12}$ **C** 9
B 3 **D** $\frac{1}{3}$

____ **4.** (1 point) In the formula for the general term of an arithmetic sequence $t_n = -7 + (n - 1) \times (-2.5)$, the common difference is
A 17.5 **C** –4.5
B –7 **D** –2.5

____ **5.** (1 point) Which of the given formulas for the general term of the sequence –9, –1, 7, 15, 23, . . . is correct?
A $t_n = -8n - 17$ **C** $t_n = 8n - 1$
B $t_n = 8n - 17$ **D** $t_n = -8n - 1$

____ **6.** (1 point) What is the 18th term of the sequence –22, –21.2, –20.4, –19.6, –18.8, …?
A –6.8 **C** –8.4
B 0.8 **D** –35.6

____ **7.** (1 point) The sum of the series (–5) + (–7) + (–9) + ··· + (–19) is
A –96 **C** –192
B –304 **D** 26

____ **8.** (1 point) The sum of an arithmetic series where $t_1 = -2$, $t_3 = 7$, and $n = 15$ is
A –502.5 **C** 476.25
B 442.5 **D** 885

____ **9.** (1 point) The sum of an arithmetic series where $t_1 = \frac{1}{2}$, $d = 3$, and $n = 19$ is
A 551 **C** $\frac{1045}{2}$
B $\frac{165}{2}$ **D** 1045

____ **10.** (1 point) For the arithmetic series (107) + (130) + (153) + ··· + (981), the values of t_1, d, and n are
A $t_1 = 107$, $d = 23$, $n = 39$ **C** $t_1 = -107$, $d = 23$, $n = 38$
B $t_1 = 107$, $d = -23$, $n = 39$ **D** $t_1 = -107$, $d = -23$, $n = 38$

____ **11.** (1 point) On the first day of the month, Michael places 5¢ in a jar. The next day, he places 7¢ in the jar. The third day, he places 9¢ in the jar, and so on for 24 days. What amount will be in the jar at the end of this period of time?

A \$6.72 **C** \$6.96
B \$6.36 **D** \$6.12

____ **12.** (1 point) The common ratio for the geometric sequence 8, 1, 0.125, 0.015625, . . . is

A $\frac{1}{8}$ **C** 8
B –8 **D** $-\frac{1}{8}$

____ **13.** (1 point) The population of a community was 82 000 at the beginning of 2000. Assuming a rate of growth of 1.6% per year since 2000, what will the population be at the beginning of 2025?

A 123 894 **C** 121 943
B 2 082 800 **D** 120 023

____ **14.** (1 point) In the formula for the general term of a geometric sequence $t_n = -10\left(\frac{4}{5}\right)^{n-1}$, the common ratio is

A $\frac{5}{4}$ **C** 4
B 5 **D** $\frac{4}{5}$

____ **15.** (1 point) The eighth term in the sequence 3 515 625, 703 125, 140 625, 28 125, … is

A 9 **C** 45
B $\frac{1}{9}$ **D** 5

____ **16.** (1 point) The 6th term in the sequence $\frac{4}{7}$, 1, $\frac{7}{4}$, $\frac{49}{16}$, . . . is

A $\frac{2401}{4096}$ **C** $\frac{256}{2401}$
B $\frac{2401}{256}$ **D** $\frac{117649}{4096}$

____ **17.** (1 point) The first three terms of the sequence given by $t_n = 11\left(\frac{1}{4}\right)^{n-1}$ are

A 11, 121, 1331 **C** $\frac{11}{4}, \frac{11}{16}, \frac{11}{64}$
B 11, $\frac{11}{4}, \frac{11}{16}$ **D** 11, $\frac{11}{3}, \frac{11}{9}$

____ **18.** (1 point) How many terms are in the sequence 2, 8, 32, 128, 512, …, 2 097 152?

A 9 **C** 10
B 12 **D** 11

____ **19.** (1 point) The sum of a geometric series where $t_1 = \frac{1}{3}$, $r = 2$, and $n = 3$ is approximately

A 2.3 **C** 2.7
B 1.3 **D** 1.2

____ **20.** (1 point) The sum of the geometric series $14 + 70 + 350 + \cdots + 43\,750$ is

A 8747 **C** 54 688

B 10 938 **D** 54 684

____ **21.** (1 point) The sum of the geometric series $13 + 6.5 + 3.25 + \cdots + 0.203125$ is

A $\frac{559}{64}$ **C** $\frac{1651}{64}$

B $\frac{1677}{64}$ **D** $\frac{819}{32}$

____ **22.** (1 point) The 6th term of the geometric series $256 + 128 + 64 + \cdots$ is

A $\frac{64}{9}$ **C** 8

B 4 **D** 2

____ **23.** (1 point) What is the value of S_9 for the series $8 - 24 + 72 - 216 + \cdots$?

A 39 366 **C** 52 491

B –13 122 **D** 39 368

____ **24.** (1 point) If you researched your ancestors back 16 generations, how many people would you need to research?

A 262 140 **C** 65 535

B 131 070 **D** 131 069

____ **25.** (1 point) Determine the sum of the infinite geometric series $11 + \frac{11}{3} + \frac{11}{9} + \frac{11}{27} + \ldots$

A 33 **C** $\frac{33}{2}$

B $\frac{33}{4}$ **D** $\frac{440}{27}$

____ **26.** (1 point) What is the sum of the infinite geometric series $15 + 15(8/9) + 15(8/9)^2 + 15(8/9)^3 + \cdots$?

A –120 **C** $\frac{135}{8}$

B $\frac{135}{17}$ **D** 135

____ **27.** (1 point) Determine the sum of the infinite geometric series with $t_1 = 2$ and $r = \frac{1}{5}$.

A $\frac{1}{3}$ **C** $-\frac{1}{2}$

B $\frac{5}{3}$ **D** $\frac{5}{2}$

____ **28.** (1 point) The sum of an infinite geometric series is $\frac{20}{3}$ and its common ratio is $\frac{1}{4}$. What is the first term of the series?

A $\frac{1}{4}$ **C** $\frac{80}{3}$

B 5 **D** $\frac{5}{3}$

____ **29.** (1 point) Which of the following best describes the series $-50 + (-45) + (-\frac{81}{2}) + (-\frac{729}{20}) + \cdots$?

A The series is convergent and has a sum of −500.
B The series is divergent and has a sum of −500.
C The series is divergent and has no sum.
D The series is convergent and has no sum.

____ **30.** (1 point) The first three terms of the sequence defined by $t_n = -0.3n + 0.5$ are

A 0.5, 0.8, 1.1
B –0.3, 0.2, 0.7
C 0.2, –0.1, –0.4
D –0.3, –0.8, –1.3

Completion
Complete each statement.

1. (1 point) In the formula for the general term of an arithmetic sequence $t_n = a + (n-1)d$, a represents the _____ and n represents the _____.

2. (1 point) 9 + 17 + 25 + 33 + 41 + ··· is an example of a(n) _____.

3. (1 point) The geometric sequence 20, 60, 180, 540, …, 393 660 has _____ terms.

4. (1 point) The geometric sequence 5, 25, 125, 625, …, 9 765 625 has _____ terms.

5. (1 point) The sum of the first 8 terms of the geometric series 5 – 15 + 45 – 135 + ··· is _____.

6. (1 point) The ratio used to generate the infinite geometric series –2.5 + 0.5 – 0.1 + 0.02 + ··· is _____.

Matching

Match the correct term to its description below.

A arithmetic series
B geometric sequence
C arithmetic sequence
D divergent series
E geometric series
F infinite geometric series

1. (1 point) a geometric series that has no last term
2. (1 point) the sum of the terms of a sequence in which the terms have a common ratio
3. (1 point) the sum of the terms of a sequence in which the terms have a common difference
4. (1 point) a sequence where there is a common difference between consecutive terms
5. (1 point) a sequence where there is a common ratio between consecutive terms

Short Answer

For each arithmetic sequence, determine
a) *the value of t_1 and d*
b) *an explicit formula for the general term*
c) t_{20}

1. (1 point) $-8, -5, -2, 1, \ldots$

2. (1 point) $3a, 3a - 2b, 3a - 4b, 3a - 6b, \ldots$

3. (1 point) $\frac{7}{4}, 1, \frac{1}{4}, -\frac{1}{2}, \ldots$

4. (1 point) The starting wage at a bookstore is \$8.50 per hour with a yearly increase of \$0.75 per hour.
a) Write the general term of the sequence representing the hourly rate earned in each year.
b) Use your expression from part a) to determine the hourly rate after 6 years.
c) How many years will someone need to work at the store to earn \$15.25 per hour?

Determine whether each sequence is geometric, arithmetic, or neither. Justify your answer.

5. (1 point) $5, -10, 20, -40, \ldots$

6. (1 point) $\frac{1}{5}, \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1, \ldots$

7. (1 point) $0.0005, 0.005, 0.05, 0.5, \ldots$

8. (1 point) $\frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \ldots$

For each geometric sequence, determine
a) *an explicit formula for the general term*
b) t_{11}

9. (1 point) $t_1 = 3, r = 2$

10. (1 point) $3, 2, \frac{4}{3}, \frac{8}{9}, \frac{16}{27}, \ldots$

11. (1 point) $3, 3\sqrt{3}, 9, 9\sqrt{3}, 27, \ldots$

12. (1 point) $-2, \frac{2}{5}, -\frac{2}{25}, \frac{2}{125}, \ldots$

For each arithmetic series, determine
a) *an explicit formula for the general term*
b) *a formula for the general sum*
c) t_{12}
d) S_n

13. (1 point) $t_1 = 2, d = 3, n = 4$

14. (1 point) $2 + 4 + 6 + \cdots + 48$

15. (1 point) $-12 - 9 - 6 - \cdots + 12$

Determine the sum of each arithmetic series.

16. (1 point) $t_1 = 3\sqrt{3},\ d = -2\sqrt{3},\ n = 11$

17. (1 point) $4k + 11k + 18k + \cdots + 74k$

18. (1 point) $(4a - 3b) + (4a + b) + (4a + 5b) + \cdots + (4a + 29b)$

19. (1 point) Find the value of t_1 given $S_8 = -3280$ and $r = -3$. Be sure to show all of your work.

20. (1 point) If $S_1 = 0.7$ and $S_2 = 2.1$ in a geometric series, determine the sum of the first 12 terms in the series. Be sure to show all of your work.

21. (1 point) A bouncy ball bounces to $\frac{2}{3}$ its height when it is dropped on a hard surface. Suppose the ball is dropped from 20 m.

a) What height will the ball bounce back up to after the sixth bounce?

b) What is the total distance the ball travels if it bounces indefinitely?

Problem

1. (1 point) Consider an arithmetic sequence with $t_2 = \frac{1}{3}$ and $t_8 = \frac{13}{3}$.

a) Determine the values of t_1 and d.

b) Determine an explicit formula for the general term of the sequence.

c) Determine the value of each term.

i) t_{12}

ii) t_{20}

iii) t_{26}

d) Determine the term number of each term.

i) $\frac{7}{3}$

ii) 9

iii) $\frac{97}{3}$

2. (1 point) In a lottery to join a golf club, the first person drawn from the names must pay \$14 000. Each subsequent person drawn pays \$250 less than the person before. The last person drawn pays \$8000 for a membership.

a) Write the first four terms of the sequence that represents the cost of a membership.

b) Determine t_1 and d for the sequence.

c) Determine an explicit formula for the general term.

d) What will the 10th golfer pay for a membership?

e) How many golfers will be able to join the club?

3. (1 point) At the end of the second week after opening, a new fitness club has 870 members. At the end of the seventh week, there are 1110 members. Suppose the increase is arithmetic.
a) How many members joined the club each week?
b) How many members were there in the first week?

4. (1 point) The sum of the first two terms of an arithmetic series is 15 and the sum of the next two terms is 43. What are the first four terms of the series?

5. (1 point) Etienne owns a small recycling company that picks up empty glass bottles from restaurants. At the first restaurant, he picks up 50 bottles. At each restaurant after this, he picks up 4 more bottles than he picked up at the restaurant before. Assume that this pattern continues.

a) Write the first four terms of the arithmetic sequence that represents the number of bottles he picks up at a restaurant.

b) Determine a and d for the sequence.

c) How many bottles will he have picked up after stopping at the eighth restaurant?

d) If his truck can hold 2000 empty bottles, will he be able to pick up bottles at the 21st restaurant without emptying the truck first?

6. (1 point) In an arithmetic series, the sum of the first 5 terms is 70 and the sum of the first 11 terms is 352.

a) Determine the values of t_1 and d.

b) What is the sum of the first

i) 30 terms?

ii) 50 terms?

iii) 100 terms?

c) Describe how you could use the formula for the sum of an arithmetic series to find the total of terms 35 to 40.

d) Use your method from part c) to determine the sum of terms 35 to 40.

7. (1 point) Consider the arithmetic series $7 + 19 + 31 + 43 + 52 + \cdots$.

a) How many terms are less than 1000?

b) How many terms are less than 2000?

c) Comment on the results from parts a) and b).

d) What is the maximum number of terms for a sum of the series less than 1000?

e) What is the maximum number of terms for a sum of the series less than 2000?

f) Comment on the results from parts d) and e).

8. (1 point) A toy car is rolling down an inclined track and picking up speed as it goes. The car travels 4 cm in the first second, 8 cm in the second second, 12 cm in the next second, and so on. What is the total distance travelled by the car in the first 30 s?

9. (1 point) A company purchases a new computer system valued at $42 000. For income tax purposes, an accountant determines that the annual depreciation rate (rate of decrease in value) for the equipment is 11%.
a) Make a table of values to show the value of the system over the first 5 years.
b) Determine an explicit formula in function notation to model the value of the system in year n.
c) What is the value of the system at the end of year 20?
d) How realistic is the answer to part c)? Explain.

10. (1 point) For each sequence, determine
i) an explicit formula for the nth term, using function notation
ii) $f(11)$
a) $1, \sqrt{3}, 3, 3\sqrt{3}, 9, 9\sqrt{3}, \ldots$
b) $5, 13, 25, 41, 61, \ldots$
c) $\frac{1}{7}, \frac{1}{3}, \frac{5}{11}, \frac{7}{13}, \frac{3}{5}, \ldots$

11. (1 point) The value of an antique increases in value at a rate of 2.5% every year. In 2000, the antique was purchased for $5000.

a) Determine an explicit formula to represent the value of the antique since the year 2000.

b) Use your formula to write the first three terms of the sequence.

c) What was the value of the antique in 2008?

d) In which year will the value of the antique be $11 866?

12. (1 point) A Registered Education Savings Plan (RESP) earns interest at a rate of 5% per year, compounded annually. Jasmine's parents invest $4000 in the account today.

a) Determine an explicit formula to represent the value of the investment.

b) Use your formula to write the first four terms of the sequence.

c) What will the investment be worth in

i) 9 years?

ii) 16 years?

d) Approximately how long will it take for the investment to double?

13. (1 point) The first prize in a lottery is \$250 000. Each winner chosen after the first is paid 20% as much as the winner before them.

a) Determine t_1 and r for the geometric sequence that represents this situation.

b) Determine an explicit formula for the general term.

c) Determine a formula for the sum of n terms of the geometric sequence.

d) If 5 winning numbers are chosen,

i) how much will the last person chosen be paid?

ii) how much will have been paid out in the lottery?

14. (1 point) A ball is kicked 26 m into the air. The ball falls, rebounds to half of its original height, and then falls to the ground again. This pattern of rebounding to half of the height from the previous bounce continues.

a) Use this information to determine t_1 and r for the geometric sequence that represents this situation.

b) Write an explicit formula for the general term.

c) Write a formula for the total distance travelled after n bounces.

d) If 7 bounces occur,

i) how high will the seventh bounce be?

ii) what total distance will the ball have travelled?

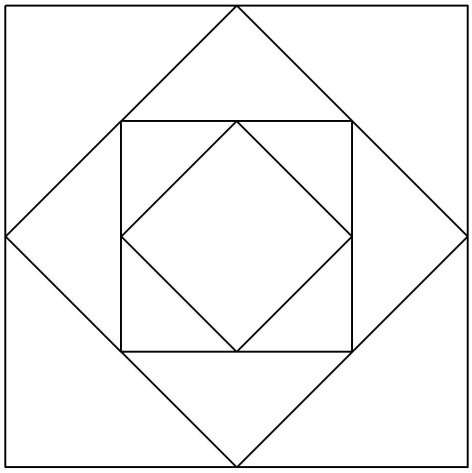
15. (1 point) In a lottery, the first ticket drawn wins a prize of $25. Each ticket after that receives a prize that is twice the value of the preceding prize.

a) Write a function to model the total amount of prize money given away.

b) How many prizes are given out if the total amount of prize money is approximately $2 million?

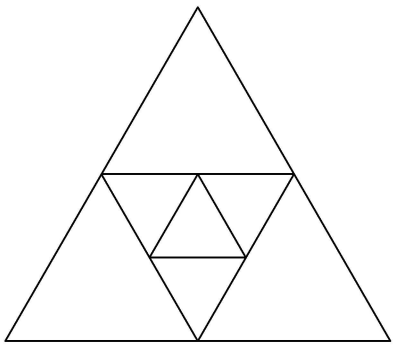
16. (1 point) For a geometric series, $\frac{S_4}{S_8} = \frac{1}{17}$. What are the first three terms of the series if the first term is 3?

17. (1 point) One side of a square is 12 cm. The midpoints of its sides are joined to form an inscribed square, and this process is continued as shown.



What is the sum of the perimeters of the squares if this process is continued without end? Give an exact answer as well as an approximation.

18. (1 point) One side of an equilateral triangle is 10 cm. The midpoints of its sides are joined to form an inscribed equilateral triangle, and this process is continued, as shown in the diagram.



What is the sum of the perimeters of the triangles if this process is continued without end?

19. (1 point) Write each repeating decimal number as an equivalent fraction in lowest terms.

a) $0.5\overline{5}55...$

b) $0.1\overline{2}$

Grade 11 Pre-Calculus: Chapter 1
Answer Section

MULTIPLE CHOICE

1. C
2. A
3. D
4. D
5. B
6. C
7. A
8. B
9. C
10. A
11. A
12. A
13. C
14. D
15. C
16. B
17. B
18. D
19. A
20. D
21. C
22. C
23. D
24. B
25. C
26. D
27. D
28. B
29. A
30. C

COMPLETION

1. a = first term
n = term number
d = common difference
2. arithmetic series
3. 10
4. 10 terms

5. –8200
6. –0.2

MATCHING

1. F
2. E
3. A
4. C
5. B

SHORT ANSWER

1. **a)** $t_1 = -8,\ d = 3$
b) $t_n = -8 + (n-1)(3)$

$= -8 + 3n - 3$

$= 3n - 11$

c) $t_{20} = 3(20) - 11$

$= 60 - 11$

$= 49$

2. **a)** $t_1 = 3a,\ d = -2b$
b) $t_n = 3a + (n-1)(-2b)$

$= 3a - 2bn + 2b$

c) $t_{20} = 3a - 2b(20) + 2b$

$= 3a - 40b + 2b$

$= 3a - 38b$

3. **a)** $t_1 = \frac{7}{4},\ d = -\frac{3}{4}$
b) $t_n = \frac{7}{4} + (n-1)\left(-\frac{3}{4}\right)$

$= \frac{7}{4} - \frac{3}{4}n + \frac{3}{4}$

$= -\frac{3}{4}n + \frac{5}{2}$

c) $t_{20} = -\frac{3}{4}(20) + \frac{5}{2}$

$= -15 + \frac{5}{2}$

$= -\frac{25}{2}$

4. **a)** $t_n = 8.50 + (n-1)(0.75)$

$= 8.50 + 0.75n - 0.75$

$= 7.75 + 0.75n$

b) $t_6 = 7.75 + 0.75(6)$

$= 12.25$

The hourly rate after 6 years is \$12.25.

c) $15.25 = 7.75 + 0.75n$

$7.50 = 0.75n$

$n = 10$

You would need to work at the bookstore for 10 years to earn \$15.25 per hour.

5. Since the terms have a common ratio of –2, this sequence is geometric.

6. Since the sequence has neither a common difference nor a common ratio, it is neither geometric nor arithmetic.

7. Since the terms have a common ratio of 10, this sequence is geometric.

8. Since the terms have a common difference of $\frac{1}{3}$, this sequence is arithmetic.

9. **a)** $t_n = 3(2)^{n-1}$

b) $t_{11} = 3(2)^{11-1}$

$= 3(2)^{10}$

$= 3072$

10. **a)** Since $t_1 = 3$ and $r = \frac{2}{3}$,

$t_n = 3\left(\frac{2}{3}\right)^{n-1}$

b) $t_{11} = 3\left(\frac{2}{3}\right)^{11-1}$

$= 3\left(\frac{2}{3}\right)^{10}$

$= \frac{1024}{19\,683}$

11. **a)** Since $t_1 = 3$ and $r = \sqrt{3}$,

$t_n = 3(\sqrt{3})^{n-1}$

b) $t_{11} = 3(\sqrt{3})^{11-1}$

$= 3(\sqrt{3})^{10}$

$= 729$

12. **a)** Since $t_1 = -2$ and $r = -\frac{1}{5}$,

$t_n = -2\left(-\frac{1}{5}\right)^{n-1}$

b) $t_{11} = -2\left(-\frac{1}{5}\right)^{11-1}$

$= -2\left(-\frac{1}{5}\right)^{10}$

$= -\frac{2}{9\,765\,625}$

13. **a)** $t_n = t_1 + (n-1)d$

$= 2 + (n-1)(3)$

$= 3n - 1$

b) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$= \frac{n}{2}[2(2) + (n-1)(3)]$

$= \frac{n}{2}(3n + 1)$

c) $t_{12} = 3(12) - 1$

$= 35$

d) $S_4 = \frac{4}{2}(3(4) + 1)$

$= 26$

14. **a)** $t_n = t_1 + (n-1)d$

$= 2 + (n-1)(2)$

$= 2n$

b) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$= \frac{n}{2}\left[2(2) + (n-1)(2)\right]$

$= \frac{n}{2}(2n+2)$

$= n^2 + n$

c) $t_{12} = 2(12)$

$= 24$

d) Since $t_n = 48$,

$48 = 2n$

$n = 24$

$S_{24} = 24^2 + 24$

$= 600$

15. **a)** $t_n = t_1 + (n-1)d$

$= -12 + (n-1)(3)$

$= 3n - 15$

b) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$= \frac{n}{2}[2(-12) + (n-1)(3)]$

$= \frac{n}{2}(3n - 27)$

c) $t_{12} = 3(12) - 15$

$= 21$

d) Since $t_n = 12$,

$12 = 3n - 15$

$3n = 27$

$n = 9$

$S_9 = \frac{9}{2}(3(9) - 27)$

$= 0$

16. $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$S_{11} = \frac{11}{2}[2(3\sqrt{3}) + (10)(-2\sqrt{3})]$

$= \frac{11}{2}(6\sqrt{3} - 20\sqrt{3})$

$= \frac{11}{2}(-14\sqrt{3})$

$= -77\sqrt{3}$

17. $t_1 = 4k,\ d = 7k$

$t_n = t_1 + (n-1)d$

$= 4k + (n-1)(7k)$

$= 4k + 7kn - 7k$

$= 7kn - 3k$

$t_n = 74k$

$74k = 7kn - 3k$

$77k = 7kn$

$n = 11$

$S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$S_{11} = \frac{11}{2}[2(4k) + (10)(7k)]$

$= \frac{11}{2}(8k + 70k)$

$= 429k$

18. $t_1 = 4a - 3b,\ d = 4b$

$t_n = t_1 + (n-1)d$

$= (4a - 3b) + (n-1)(4b)$

$= 4a - 3b + 4nb - 4b$

$= 4a - 7b + 4nb$

$t_n = 4a + 29b$

$4a + 29b = 4a - 7b + 4nb$

$36b = 4nb$

$n = 9$

$S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$S_9 = \frac{9}{2}[2(4a - 3b) + (8)(4b)]$

$= \frac{9}{2}(8a - 6b + 32b)$

$= \frac{9}{2}(8a + 26b)$

$= 9(4a + 13b)$

$= 36a + 117b$

19. $S_n = \frac{t_1(r^n - 1)}{r - 1}$

$-3280 = \frac{t_1[(-3)^8 - 1]}{-3 - 1}$

$-3280 = \frac{t_1(6561 - 1)}{-4}$

$13\ 120 = t_1(6560)$

$t_1 = 2$

20. In the series, $t_1 = 0.7$. To find t_2, evaluate $S_2 - S_1$.

$t_2 = S_2 - S_1$

$= 2.1 - 0.7$

$= 1.4$

To find r, evaluate $\frac{t_2}{t_1}$.

$r = \frac{t_2}{t_1}$

$= \frac{1.4}{0.7}$

$= 2$

So, $r = 2$.

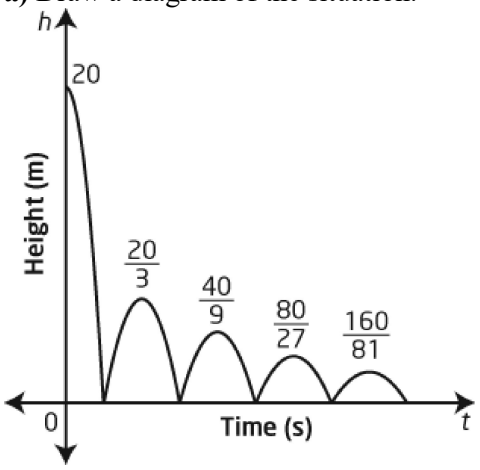
$S_n = \frac{t_1(r^n - 1)}{r - 1}$

$S_{12} = \frac{0.7(2^{12} - 1)}{2 - 1}$

$= \frac{0.7(4096 - 1)}{1}$

$= 2866.5$

21. **a)** Draw a diagram of the situation.



$t_1 = 20\left(\frac{2}{3}\right)$ and $r = \frac{2}{3}$

$= \frac{40}{3}$

$t_n = t_1 r^{n-1}$

$t_6 = \frac{40}{3}\left(\frac{2}{3}\right)^5$

≈ 1.76

The ball will bounce to a height of approximately 1.76 m on the sixth bounce.

b) The distance the ball travels (both down and up) is
$20 + 2t_1 + 2t_2 + 2t_3 + \cdots = 20 + 2S_\infty$.

$20 + 2S_\infty = 20 + 2\left(\frac{t_1}{1-r}\right)$

$= 20 + 2\left(\frac{\frac{40}{3}}{1-\frac{2}{3}}\right)$

$= 20 + 2\left(\frac{\frac{40}{3}}{\frac{1}{3}}\right)$

$= 20 + 2\left(\frac{40}{3}\right)\left(\frac{3}{1}\right)$

$= 20 + 80$

$= 100$

The total distance the ball will travel is 100 m.

PROBLEM

1. **a)** Use $t_n = a + (n-1)d$ to write and solve a system of equations.

$t_2 = t_1 + (2-1)d$ and $t_8 = t_1 + (8-1)d$

$\frac{1}{3} = t_1 + d$ (1) $\frac{13}{3} = t_1 + 7d$ (2)

$\frac{13}{3} = t_1 + 7d$ (2)

$\frac{1}{3} = t_1 + d$ (1)

$\frac{12}{3} = 6d$ (2) – (1)

$4 = 6d$

$d = \frac{2}{3}$

Substitute $d = \frac{2}{3}$ into one of the original equations in the system.

$\frac{1}{3} = t_1 + \frac{2}{3}$

$t_1 = -\frac{1}{3}$

b) $t_n = t_1 + (n-1)d$

$= -\frac{1}{3} + (n-1)\left(\frac{2}{3}\right)$

$= -\frac{1}{3} + \frac{2}{3}n - \frac{2}{3}$

$= \frac{2}{3}n - 1$

c) i) $t_{12} = \frac{2}{3}(12) - 1$

$= 8 - 1$

$= 7$

ii) $t_{20} = \frac{2}{3}(20) - 1$

$= \frac{40}{3} - 1$

$= \frac{37}{3}$

iii) $t_{26} = \frac{2}{3}(26) - 1$

$= \frac{52}{3} - 1$

$= \frac{49}{3}$

d) i) $\frac{7}{3} = \frac{2}{3}n - 1$

$\frac{10}{3} = \frac{2}{3}n$

$n = 5$

ii) $9 = \frac{2}{3}n - 1$

$9 + 1 = \frac{2}{3}n$

$10 = \frac{2}{3}n$

$n = 15$

iii) $\frac{97}{3} = \frac{2}{3}n - 1$

$\frac{100}{3} = \frac{2}{3}n$

$n = 50$

2. **a)** 14 000, 13 750, 13 500, 13 250

b) $t_1 = 14\,000$, $d = -250$

c) $t_n = t_1 + (n-1)d$

$= 14\,000 + (n-1)(-250)$

$= 14\,000 - 250n + 250$

$= 14\,250 - 250n$

d) $t_{10} = 14\,250 - 250(10)$

$= 11\,750$

Therefore, the 10th golfer will pay \$11 750.

e) The last person pays \$8000, so $t_n = 8000$.

$t_n = 14\,250 - 250n$

$8000 = 14\,250 - 250n$

$-6250 = -250n$

$n = 25$

Therefore, 25 golfers will be able to join the club.

3. **a)** $t_n = t_1 + (n-1)d$

$t_2 = 870$

$t_7 = 1110$

$t_7 - t_2 = (7-1)d - (2-1)d$

$= 5d$

$1110 - 870 = 5d$

$240 = 5d$

$d = 48$

Each week, 48 members joined the club.

b) $t_2 = t_1 + d$

$870 = t_1 + 48$

$t_1 = 822$

There were 822 members in the first week.

4. $t_1 + t_2 = 15$ $\qquad$ $t_3 + t_4 = 43$

$t_1 + (t_1 + d) = 15$ $\qquad$ $(t_1 + 2d) + (t_1 + 3d) = 43$

$2t_1 + d = 15$ $\qquad$ $2t_1 + 5d = 43$

Solve the system of equations to determine t_1 and d.

$2t_1 + d = 15$

$2t_1 + 5d = 43$

$-4d = -28$

$d = 7$

Substitute $d = 7$ into the first equation to solve for t_1.

$2t_1 + d = 15$

$2t_1 + 7 = 15$

$2t_1 = 8$

$t_1 = 4$

The first four terms are 4, 11, 18, and 25.

5. **a)** 50, 54, 58, 62

b) $t_1 = 50$, $d = 4$

c) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$S_8 = \frac{8}{2}[2(50) + (8-1)(4)]$

$= 4(100 + 28)$

$= 512$

He will have picked up 512 bottles after stopping at the eighth restaurant.

d) $S_{20} = \frac{20}{2}[2(50) + (20-1)(4)]$

$= 10(100 + 76)$

$= 1760$

Use the explicit formula for the general term of an arithmetic sequence to determine the number of bottles at the 21st restaurant.

$t_n = t_1 + (n-1)d$

$t_{21} = 50 + (21-1)(4)$

$= 50 + (20)(4)$

$= 50 + 80$

$= 130$

He has 1760 bottles in the truck and will be picking up 130 more at the next restaurant. This would bring the total to 1760 + 130, or 1890, which is less than 2000. This means that he can stop at the next restaurant.

6. **a)** Use $S_n = \frac{n}{2}[2t_1 + (n-1)d]$ to write and solve a system of equations.

$S_5 = \frac{5}{2}[2t_1 + (5-1)d]$ $\qquad$ $S_{11} = \frac{11}{2}[2t_1 + (11-1)d]$

$70 = \frac{5}{2}[2t_1 + 4d]$ $\qquad$ $352 = \frac{11}{2}[2t_1 + 10d]$

$70 = 5t_1 + 10d$ $\qquad$ $352 = 11t_1 + 55d$

$14 = t_1 + 2d$ $\qquad$ $32 = t_1 + 5d$

Solve the system of equations.

$32 = t_1 + 5d$

$\underline{14 = t_1 + 2d}$

$18 = 3d$

$d = 6$

Substitute $d = 6$ into one of the equations.

$14 = t_1 + 2d$

$14 = t_1 + 2(6)$

$t_1 = 2$

$t_1 = 2$, $d = 6$

b) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$= \frac{n}{2}[2(2) + (n-1)6]$

$= \frac{n}{2}(4 + 6n - 6)$

$= \frac{n}{2}(6n - 2)$

$= n(3n - 1)$

i) $S_{30} = 30(3(30) - 1)$
$= 30(90 - 1)$
$= 30(89)$
$= 2670$

ii) $S_{50} = 50(3(50) - 1)$
$= 50(150 - 1)$
$= 50(149)$
$= 7450$

iii) $S_{100} = 100(3(100) - 1)$
$= 100(300 - 1)$
$= 100(299)$
$= 29\,900$

c) Find the sum of the first 34 terms and the sum of the first 40 terms and subtract the two results.

d) $S_{40} = 40(3(40) - 1)$ and $S_{34} = 34(3(34) - 1)$

$= 40(120 - 1)$ $\qquad$ $= 34(102 - 1)$

$= 40(119)$ $\qquad$ $= 34(101)$

$= 4760$ $\qquad$ $= 3434$

Therefore, the required sum is 4760 – 3434, or 1326.

7. **a)** $t_n = t_1 + (n-1)d$

$= 7 + (n-1)(12)$

$= 12n - 5$

For terms less than 1000:

$1000 = 12n - 5$

$1005 = 12n$

$n = 83.75$

Therefore, 83 terms are less than 1000.

b) For terms less than 2000:

$2000 = 12n - 5$

$2005 = 12n$

$n \doteq 167.08$

Therefore, 167 terms are less than 2000.

c) Answers may vary. Sample answer: There are approximately twice as many terms less than 2000 as there are terms less than 1000. This makes sense because the common difference of 12 increases terms from 0 to 1000 in the same way as it does from 1000 to 2000.

d) $S_n = \frac{n}{2}[2t_1 + (n-1)d]$

$= \frac{n}{2}[2(7) + (n-1)(12)]$

$= \frac{n}{2}[14 + 12n - 12]$

$= \frac{n}{2}[12n + 2]$

$= 6n^2 + n$

For a sum less than 1000:

$1000 = 6n^2 + n$

$6n^2 + n - 1000 = 0$

Using the quadratic formula, the positive solution (ignore the negative solution) is $n \approx 12.83$. So, 12 is the maximum number of terms with a sum less than 1000.

e) For a sum less than 2000:

$2000 = 6n^2 + n$

$6n^2 + n - 2000 = 0$

Using the quadratic formula, the positive solution (ignore the negative solution) is $n \approx 18.17$. So, 18 is the maximum number of terms with a sum less than 2000.

f) Answers may vary. Sample answer: Unlike the number of terms from parts a) and b), the maximum number of terms for the sums does not double from part d) to part e). This is because as the difference is applied to terms, they continue to get larger and larger, which affects the sum of terms dramatically.

8. $t_1 = 4$ and $d = 4$.

$$S_n = \frac{n}{2}[2t_1 + (n-1)d]$$

$$S_{30} = \frac{30}{2}[2(4) + (30-1)4]$$

$$= 15[8 + (29)4]$$

$$= 15[124]$$

$$= 1860$$

The toy car travels 1860 cm or 18.6 m.

9. a)

Year	Value ($)
0	42 000.00
1	37 380.00
2	33 268.20
3	29 608.70
4	26 351.74
5	23 453.05

b) $V(n) = 42\,000(0.89)^n$, where V represents the value and n represents the number of years since the system was purchased.

c) $V(20) = 42\,000(0.89)^{20}$

$$\doteq 4083.66$$

The value of the system after 20 years is $4083.66.

d) This value is most likely not realistic, as most companies would replace a computer system before the 20-year mark to upgrade the system to current needs. This would often be done before the end of the 10th year of owning a new system (or less, for some companies that require more up-to-date computer technology).

10. **a) i)** $f(n) = f(1)r^{n-1}$

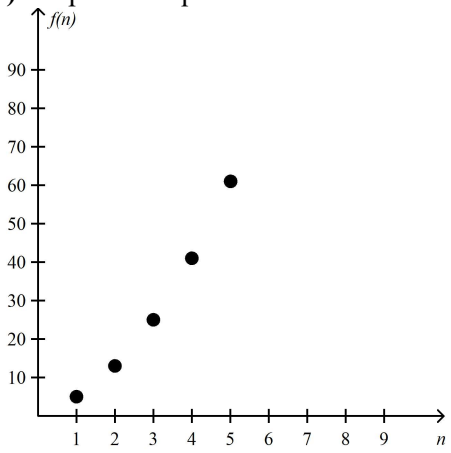
$= (\sqrt{3})^{n-1}$

ii) $f(11) = (\sqrt{3})^{11-1}$

$= (\sqrt{3})^{10}$

$= 243$

b) i) Graph the sequence on a coordinate grid with n on the horizontal axis and t_n on the vertical axis.



The points look like they lie on a parabola. Thus, the formula for the nth term is of the form $f(n) = an^2 + bn + c$.

Substitute $f(1)$, $f(2)$, and $f(3)$ into this formula and solve for a, b, and c.

$5 = a(1)^2 + b(1) + c$ $\quad 13 = a(2)^2 + b(2) + c$ $\quad 25 = a(3)^2 + b(3) + c$

$5 = a + b + c$ (1) $\quad 13 = 4a + 2b + c$ (2) $\quad 25 = 9a^2 + 3b + c$ (3)

Subtract (1) from (2):

$13 = 4a + 2b + c$

$\underline{5 = a + b + c}$

$8 = 3a + b$ (4)

Subtract (2) from (3):

$25 = 9a + 3b + c$

$\underline{13 = 4a + 2b + c}$

$12 = 5a + b$ (5)

Subtract (4) from (5):

$12 = 5a + b$

$\underline{8 = 3a + b}$

$4 = 2a$

$a = 2$

Substituting $a = 2$ into $8 = 3a + b$ gives $b = 2$.

Substituting $a = 2$ and $b = 2$ into $a + b + c = 5$ gives $c = 1$.

Thus, the formula for the nth term is $f(n) = 2n^2 + 2n + 1$.

ii) $f(11) = 2(11)^2 + 2(11) + 1$

$= 242 + 22 + 1$

$= 265$

c) i) The denominators of t_1, t_3, and t_4 look like part of the sequence 7, 9, 11, 13, 15, ... Rewriting t_2 with a denominator of 9 gives $\frac{3}{9}$, and rewriting t_5 with a denominator of 15 gives $\frac{9}{15}$. Thus, the sequence is

$\frac{1}{7}, \frac{3}{9}, \frac{5}{11}, \frac{7}{13}, \frac{9}{15}, \ldots$.

The numerator is an arithmetic sequence with $a = 1$ and $d = 2$, and the denominator is an arithmetic sequence with $a = 7$ and $d = 2$.

Thus,

$$f(n) = \frac{1 + (n-1)2}{7 + (n-1)2}$$

$$= \frac{1 + 2n - 2}{7 + 2n - 2}$$

$$= \frac{2n - 1}{2n + 5}$$

ii) $f(11) = \frac{2(11) - 1}{2(11) + 5}$

$$= \frac{22 - 1}{22 + 5}$$

$$= \frac{21}{27}$$

$$= \frac{7}{9}$$

11. **a)** At a rate of increase of 2.5% per year, $r = 1.025$. The initial value of the antique in 2000 is \$5000, so $t_0 = 5000$.

$t_n = 5000(1.025)^n$

b) $t_1 = 5000(1.025)^1$ $\quad$ $t_2 = 5000(1.025)^2$ $\quad$ $t_3 = 5000(1.025)^3$

$= 5125$ $\quad$ ≈ 5253.13 $\quad$ ≈ 5384.45

The first three terms of the sequence are \$5125, \$5253.13, and \$5384.45.

c) The year 2008 represents t_8.

$t_8 = 5000(1.025)^8$

$\doteq 6092.01$

d) $11\,866 = 5000(1.025)^n$

$2.3732 = (1.025)^n$

Using systematic trial, $(1.025)^{35} \approx 2.3732$. So, $n = 35$. In 2035, the antique will be worth approximately \$11 866.

12. **a)** At an interest rate of 5% per year, $r = 1.05$. The initial value of the investment is \$4000, so $t_0 = 4000$.
$t_n = 4000(1.05)^n$

b) $t_1 = 4000(1.05)^1$ $t_2 = 4000(1.05)^2$ $t_3 = 4000(1.05)^3$ $t_4 = 4000(1.05)^4$

$= 4200$ $= 4410$ $= 4630.50$ $= 4862.02$

c) i) $t_9 = 4000(1.05)^9$ **ii)** $t_{16} = 4000(1.05)^{16}$

$\doteq 6205.31$ ≈ 8731.50

d) $8000 = 4000(1.05)^n$

$2 = (1.05)^n$

Using systematic trial, $(1.05)^{14} \doteq 1.9799$. So, $n = 14$. In approximately 14 years, the investment will double.

13. **a)** $t_1 = 250\,000$, $r = 0.2$

b) $t_n = 250\,000(0.2)^{n-1}$

c) $S_n = \dfrac{t_1(r^n - 1)}{r - 1}$

$= \dfrac{250\,000(0.2^n - 1)}{0.2 - 1}$

$= \dfrac{250\,000(0.2^n - 1)}{-0.8}$

$= -312\,500(0.2^n - 1)$

d) i) $t_5 = 250\,000(0.2)^{5-1}$

$= 400$

Therefore, the 5th person will win \$400.

ii) $S_5 = -312\,500(0.2^5 - 1)$

$= 312\,400$

Therefore, \$312 400 will have been paid out in the lottery.

14. **a)** $t_1 = 26, r = 0.5$

b) $t_n = 26(0.5)^{n-1}$

c) You need to double the height to find the total distance travelled, as the ball must bounce up to the height and then back to the ground.

$$2S_n = 2\left(\frac{t_1(r^n - 1)}{r - 1}\right)$$

$$= \frac{2(26)(0.5^n - 1)}{0.5 - 1}$$

$$= \frac{52(0.5^n - 1)}{-0.5}$$

$$= -104(0.5^n - 1)$$

d) i) $t_7 = 26(0.5)^{7-1}$

$$= 0.406\ 25$$

Therefore, the seventh bounce will be approximately 0.41 m high.

ii) $2S_7 = \frac{52(0.5^7 - 1)}{-0.5}$

$$= 103.1875$$

Therefore, the ball will have travelled approximately 103 m after the seventh bounce.

15. **a)** $t_1 = 25$ and $r = 2$ Therefore,

$$S_n = \frac{25(2^n - 1)}{2 - 1}$$

$$= 25(2^n - 1)$$

b) $2\ 000\ 000 = 25(2^n - 1)$

$$80\ 000 = 2^n - 1$$

$$2^n = 80\ 001$$

Using trial and error, $2^{16} = 65\ 536$ and $2^{17} = 131\ 072$, so 16 prizes were handed out.

16. $\dfrac{S_4}{S_8} = \dfrac{1}{17}$

$$\frac{\dfrac{t_1(r^4-1)}{r-1}}{\dfrac{t_1(r^8-1)}{r-1}} = \frac{1}{17}$$

$$\frac{r^4-1}{r^8-1} = \frac{1}{17}$$

$17r^4 - 17 = r^8 - 1$

$r^8 - 17r^4 + 16 = 0$

$(r^4-1)(r^4-16) = 0$

$r^4 = 1$ or $r^4 = 16$

$r = \pm 1$ or $r = \pm 2$

From the definition of a geometric series, $r \neq \pm 1$, so the first three terms are 3, 6, 12 or 3, –6, 12.

17. Let x_n represent the side length of the nth square.
$x_1 = 12$
Let P_n represent the perimeter of the nth square.
$P_1 = 48$
Use the Pythagorean relation to find x_2.

$$(x_2)^2 = \left(\frac{x_1}{2}\right)^2 + \left(\frac{x_1}{2}\right)^2$$
$$= 6^2 + 6^2$$
$$= 72$$
$$x_2 = \sqrt{72}$$
$$= 6\sqrt{2}$$

$P_2 = 24\sqrt{2}$
Find x_3 and P_3 in the same way:

$$(x_3)^2 = \left(\frac{x_2}{2}\right)^2 + \left(\frac{x_2}{2}\right)^2$$
$$= (3\sqrt{2})^2 + (3\sqrt{2})^2$$
$$= 36$$
$$x_3 = \sqrt{36}$$
$$= 6$$

$P_3 = 24$
Use these results to find r in the geometric series for P_n.

$$r = \frac{P_2}{P_1} \quad \text{and} \quad r = \frac{P_3}{P_2}$$
$$= \frac{24\sqrt{2}}{48} \qquad = \frac{24}{24\sqrt{2}}$$
$$= \frac{\sqrt{2}}{2} \qquad = \frac{1}{\sqrt{2}}$$
$$= \frac{\sqrt{2}}{2}$$

The common ratio, r, is $\frac{\sqrt{2}}{2}$. Use the formula for the sum of an infinite geometric series:

$$P_\infty = \frac{t_1}{1-r}$$

$$= \frac{48}{1-\frac{\sqrt{2}}{2}}$$

$$= \frac{48}{\frac{2-\sqrt{2}}{2}}$$

$$= \frac{96}{2-\sqrt{2}}$$

$$= \frac{96}{2-\sqrt{2}}\left(\frac{2+\sqrt{2}}{2+\sqrt{2}}\right)$$

$$= \frac{192+96\sqrt{2}}{4-2}$$

$$= \frac{96(2+\sqrt{2})}{2}$$

$$= 48(2+\sqrt{2})$$

$$\approx 163.9$$

The total perimeter is $48(2+\sqrt{2})$ cm or approximately 163.9 cm.

18. The first perimeter, t_1, is 30 cm.

The common ratio, r, is $\frac{1}{2}$.

Use the formula for the sum of an infinite geometric series.

$$S_\infty = \frac{t_1}{1-r}$$

$$= \frac{30}{1-\frac{1}{2}}$$

$$= \frac{30}{\frac{1}{2}}$$

$$= 60$$

The total perimeter is 60 cm.

19. **a)** $0.\overline{5} = 0.5 + 0.05 + 0.005 + \cdots$
This is an infinite geometric series with $t_1 = 0.5$ and $r = 0.1$.

$$S_\infty = \frac{t_1}{1-r}$$

$$= \frac{0.5}{1-0.1}$$

$$= \frac{0.5}{0.9}$$

$$= \frac{5}{9}$$

b) $0.1\overline{2} = 0.1 + 0.02 + 0.002 + 0.0002 + \cdots$
After 0.1, $t_1 = 0.02$ and $r = 0.1$.

$$S_\infty = \frac{t_1}{1-r}$$

$$= \frac{0.02}{1-0.1}$$

$$= \frac{0.02}{0.9}$$

$$= \frac{2}{90}$$

Add this fraction to 0.1 (or $\frac{1}{10}$):

$$\frac{1}{10} + \frac{2}{90} = \frac{9+2}{90}$$

$$= \frac{11}{90}$$